Masked observations

1 Defining equations

Lemma 1.1. *For every natural number Q and natural number t,* $\mathrm{centergate}(Q, t) = \mathrm{not}(\mathrm{center}(t))\,\mathrm{and}\,\mathrm{rightdefect}(Q, t, 1)$.

Proof. The goal follows from $\mathrm{centergatedefinition}(Q, t)$. □

Lemma 1.2. *For every natural number Q and natural number t,* $\mathrm{alignedbackground}(Q, t) = \mathrm{wittshl}(\mathrm{wittrow}(t), Q)$.

Proof. The goal follows from $\mathrm{backgrounddefinition}(Q, t)$. □

Lemma 1.3. *For every natural number Q and natural number t,* $\mathrm{aligneddefect}(Q, t) = \mathrm{wittxor}(\mathrm{wittrow}(t + Q), \mathrm{alignedbackground}(Q, t))$.

Proof. The goal follows from $\mathrm{aligneddefectdefinition}(Q, t)$. □

Lemma 1.4. *For every natural number L and natural number H,* $\mathrm{carryledger}(L, H, 0) = 0$.

Proof. The goal follows from $\mathrm{carryledgerdefinition}(L, H)$. □

Lemma 1.5. *For every natural number L and natural number H and natural number t,* $\mathrm{carryledger}(L, H, \mathrm{succ}(t)) = \mathrm{wittxor}(\mathrm{carryledger}(L, H, t), \mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{page}(L, H, t)), \mathrm{na}$

Proof. The goal follows from $\mathrm{carryledgerdefinition2}(L, H, t)$. □

Lemma 1.6. *For every natural number n,* $\mathrm{centerflipmask}(n) = \mathrm{wittshl}(7, n)$.

Proof. The goal follows from $\mathrm{centerflipmaskdefinition}(n)$. □

Lemma 1.7. *For every natural number $start$,* $\mathrm{centersign}(start, 0) = 1$.

Proof. The goal follows from $\mathrm{centersigndefinition}(start)$. □

Lemma 1.8. *For every natural number $start$ and natural number t,* $\mathrm{centersign}(start, \mathrm{succ}(t)) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(start))) \cdot \mathrm{centersign}(start + 1, t)$.

Proof. The goal follows from $\mathrm{centersigndefinition2}(start, t)$. □

Lemma 1.9. *For every natural number b and natural number d,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittphi}(d), \mathrm{wittshl}(\mathrm{crosscarry}(b, d), 1))$.

Proof. The goal follows from $\mathrm{defectstepdefinition}(b, d)$. □

Lemma 1.10. *For every natural number Q and natural number d and natural number t,* diagonalphase(Q, d, t) = bitvalue(ldiag$(d, t{+}Q)$)$-$bitvalue(ldiag(d, t)).

Proof. The goal follows from diagonalphasedefinition(Q, d, t). □

Lemma 1.11. *For every natural number Q and natural number t,* observersourceflux$(Q, t, 0)$ = 0.

Proof. The goal follows from observersourcefluxdefinition(Q, t). □

Lemma 1.12. *For every natural number Q and natural number t and natural number $t1$,* observersourceflux$(Q, t, \text{succ}(t1))$ = wittxor(wittlinear(observersourceflux$(Q, t, t1)$), wittshl($t1), t1 + 1)$).

Proof. The goal follows from observersourcefluxdefinition2$(Q, t, t1)$. □

Lemma 1.13. *For every natural number L and natural number H,* pageboundarycarry$(L, H, 0)$ = false.

Proof. The goal follows from pageboundarycarrydefinition(L, H). □

Lemma 1.14. *For every natural number L and natural number H and natural number t,* pageboundarycarry$(L, H, \text{succ}(t))$ = rowbit(page$(L, H, t + 1), 0)$ and rowbit(page$(L, H, t), L - 1)$.

Proof. The goal follows from pageboundarycarrydefinition2(L, H, t). □

Lemma 1.15. *For every natural number L and natural number H,* pagefold$(L, H, 0)$ = 0.

Proof. The goal follows from pagefolddefinition(L, H). □

Lemma 1.16. *For every natural number L and natural number H and natural number t,* pagefold$(L, H, \text{succ}(t))$ = wittxor(pagefold(L, H, t), page(L, H, t)).

Proof. The goal follows from pagefolddefinition2(L, H, t). □

Lemma 1.17. *For every natural number Q and natural number t and natural number k,* pathflux$(Q, t, k, 0) = 0$.

Proof. The goal follows from pathfluxdefinition(Q, t, k). □

Lemma 1.18. *For every natural number d,* qordersearch$(d, 0) = 0$.

Proof. The goal follows from qordersearchdefinition(d). □

Lemma 1.19. *For every natural number t and natural number k,* $\mathrm{rayflux}(t, k, 0) = 0$.

Proof. The goal follows from $\mathrm{rayfluxdefinition}(t, k)$. □

Lemma 1.20. *For every truth value a and truth value g and integer $outgoing$ and integer $gate$,* $\mathrm{recoverphase}(a, g, outgoing, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot (outgoing - (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate)$.

Proof. The goal follows from $\mathrm{recoverphasedefinition}(a, g, outgoing, gate)$. □

Lemma 1.21. *For every natural number t,* $\mathrm{rightimport}(t) = \mathrm{not}(\mathrm{rightcell}(t, 1))\,\mathrm{and}\,\mathrm{rightcell}(t, 2)$.

Proof. The goal follows from $\mathrm{rightimportdefinition}(t)$. □

Lemma 1.22. *For every natural number Q and natural number t and natural number k,* $\mathrm{rightphase}(Q, t, k) = \mathrm{bitvalue}(\mathrm{rightcell}(t{+}Q, k)){-}\mathrm{bitvalue}(\mathrm{rightcell}(t, k))$.

Proof. The goal follows from $\mathrm{rightphasedefinition}(Q, t, k)$. □

Lemma 1.23. *For every truth value a and truth value g and integer $incoming$ and integer $gate$,* $\mathrm{transferphase}(a, g, incoming, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot incoming + (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate$.

Proof. The goal follows from $\mathrm{transferphasedefinition}(a, g, incoming, gate)$. □

Lemma 1.24. *For every natural number n,* $\mathrm{twopow}(\mathrm{succ}(n)) = \mathrm{twopow}(n){+}\,\mathrm{twopow}(n)$.

Proof. The goal follows from $\mathrm{twopowdefinition2}(n)$. □

Lemma 1.25. *For every natural number y,* $\mathrm{wittlinear}(y) = \mathrm{wittxor}(\mathrm{wittxor}(y, \mathrm{wittshl}(y, 1)), \mathrm{wittshl}(y, 2$

Proof. The goal follows from $\mathrm{philineardefinition}(y)$. □

Lemma 1.26. *For every natural number $h1$ and natural number m,* $\mathrm{wittlinpow}(\mathrm{succ}(m), h1) = \mathrm{wittlinear}(\mathrm{wittlinpow}(m, h1))$.

Proof. The goal follows from $\mathrm{philinearpowdefinition2}(h1, m)$. □

Lemma 1.27. *For every natural number n,* $\mathrm{wittrow}(\mathrm{succ}(n)) = \mathrm{wittphi}(\mathrm{wittrow}(n))$.

Proof. The goal follows from $\mathrm{rowdefinition2}(n)$. □

Lemma 1.28. *For every natural number Q and natural number t and natural number i,* $\mathrm{alignedphase}(Q, t, i) = \mathrm{pairphase}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t), i)$.

Proof. The goal follows from alignedphasedefinition(Q, t, i). □

Lemma 1.29. *For every natural number h and natural number A,* blockendpoint$(h, A) =$ rowbit$(A, 0)$ and rowbit$(A, h - 1)$.

Proof. The goal follows from blockendpointdefinition(h, A). □

Lemma 1.30. *For every natural number t,* center$(t) =$ cell$(t, 0)$.

Proof. The goal follows from centerdefinition(t). □

Lemma 1.31. *For every natural number Q and natural number d and natural number t and natural number p,* diagonalflux$(Q, d, t, p) =$ rayflux$(t + Q, t + Q - d, t + Q - d - p) -$ rayflux$(t, t - d, t - d - p)$.

Proof. The goal follows from diagonalfluxdefinition(Q, d, t, p). □

Lemma 1.32. *For every natural number Q and natural number t and natural number k,* erasureevent$(Q, t, k) =$ (leftdefect$(Q, t, k+1) =$ true $\wedge$ leftdefect$(Q, t+1, k) =$ false $\wedge$ gatecode$(Q, t, k) \neq 0 \wedge$collisioncode$(Q, t, k) = 0 \wedge$(leftcolour$(Q, t+1, k) = 0 \vee$ leftcolour$(Q, t + 1, k) = 3)$).

Proof. The goal follows from erasureeventdefinition(Q, t, k). □

Lemma 1.33. ldiag$(0, 0) =$ true.

Proof. The goal follows from ldiagdefinition. □

Lemma 1.34. *For every natural number n,* ldiag$(\text{succ}(n), 0) =$ false.

Proof. The goal follows from ldiagdefinition2(n). □

Lemma 1.35. *For every natural number s,* ldiag$(0, \text{succ}(s)) =$ true.

Proof. The goal follows from ldiagdefinition3(s). □

Lemma 1.36. *For every natural number s,* ldiag$(\text{succ}(0), \text{succ}(s)) =$ true.

Proof. The goal follows from ldiagdefinition4(s). □

Lemma 1.37. *For every natural number s and natural number $n1$,* ldiag$(\text{succ}(\text{succ}(n1)), \text{succ}(s)) =$ bxor(ldiag$(n1, s)$, ldiag$(\text{succ}(n1), s)$ or ldiag$(\text{succ}(\text{succ}(n1)), s)$).

Proof. The goal follows from ldiagdefinition5$(s, n1)$. □

Lemma 1.38. *For every natural number Q and natural number t and natural number k,* $\mathrm{leadingadvance}(Q, t, k) = (2 \leq k \wedge \mathrm{advancingatdepth}(Q, t, k) \wedge \mathrm{rightmostleft}(Q, t, k))$.

Proof. The goal follows from $\mathrm{leadingadvancedefinition}(Q, t, k)$. □

Lemma 1.39. *For every truth value b,* $\mathrm{naturalbit}(b) = \mathrm{naturalbit}(b)$.

Proof. The goal follows from $\mathrm{naturalbitdefinition}(b)$. □

Lemma 1.40. *For every natural number t,* $\mathrm{observerinput}(t) = \mathrm{not}(\mathrm{center}(t))$ and $\mathrm{rightcell}(t, 1)$.

Proof. The goal follows from $\mathrm{observerinputdefinition}(t)$. □

Lemma 1.41. *For every natural number Q and natural number t,* $\mathrm{observerqrank}(Q, t) = t + Q - 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t))$.

Proof. The goal follows from $\mathrm{observerqrankdefinition}(Q, t)$. □

Lemma 1.42. *For every natural number b and natural number d and natural number i,* $\mathrm{pairphase}(b, d, i) = \mathrm{bitvalue}(\mathrm{rowbit}(\mathrm{wittxor}(b, d), i)) - \mathrm{bitvalue}(\mathrm{rowbit}(b, i))$.

Proof. The goal follows from $\mathrm{pairphasedefinition}(b, d, i)$. □

Lemma 1.43. *For every natural number a and natural number i,* $\mathrm{rowbit}(a, i) = \mathrm{rowbit}(a, i)$.

Proof. The goal follows from $\mathrm{rowbitdefinition}(a, i)$. □

Lemma 1.44. *For every natural number $h1$,* $\mathrm{wittwin}(h1, 0) = 0$.

Proof. The goal follows from $\mathrm{winwittdefinition}(h1)$. □

Lemma 1.45. *For every natural number $h1$ and natural number s,* $\mathrm{wittwin}(h1, \mathrm{succ}(s)) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{wittwin}(h1, s)), \mathrm{wittshl}(\mathrm{wittcarry}(\mathrm{wittrow}(h1 + s)), 1))$.

Proof. The goal follows from $\mathrm{winwittdefinition2}(h1, s)$. □

2 Masked observations

Theorem 2.1. *For every natural number t,* $\mathrm{whitecount}(t, 0) = 0$.

Proof. The goal follows from $\mathrm{whitecountzero}(t)$. □

Theorem 2.2. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n)$.

Proof. The goal follows from $\mathrm{whitecountblackstep}(t, n)$. □

Theorem 2.3. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n) + 1$.

Proof. The goal follows from $\mathrm{whitecountwhitestep}(t, n)$. □

Theorem 2.4. *For every natural number $t1$ and natural number $t2$ and natural number n, if* $\mathrm{centerwindowagree}(t1, t2, n)$*, then* $\mathrm{leftprefixagree}(t1, t2, n)$ *if and only if* $\mathrm{whitetraceagree}(t1, t2, n)$.

Proof. The goal follows from $\mathrm{prefixagreementiffwhitetrace}(t1, t2, n)$. □

Theorem 2.5. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then there exists a natural number a such that there exists a natural number b such that $a < b$ and $b \leq \mathrm{twopow}(\mathrm{whitecount}(t, n)) \wedge \mathrm{leftprefixagree}(t + a \cdot P, t + b \cdot P, n)$.*

Proof. The goal follows from $\mathrm{compressedprefixrepeat}(T, P, t, n)$. □

Theorem 2.6. *For every natural number T and natural number P and natural number t and natural number n and natural number M, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then if* $\mathrm{prefixseparated}(t, P, n, M)$*, then $M \leq \mathrm{twopow}(\mathrm{whitecount}(t, n))$.*

Proof. The goal follows from $\mathrm{separatedprefixbound}(T, P, t, n, M)$. □

Theorem 2.7. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then $n < t + \mathrm{twopow}(\mathrm{whitecount}(t, n)) \cdot P$.*

Proof. The goal follows from $\mathrm{edgeinformationbound}(T, P, t, n)$. □

Theorem 2.8. *For every natural number n and natural number t and natural number w and natural number P, if $n < (w+1) \cdot P$, then $n < t + \mathrm{twopow}(w) \cdot P$.*

Proof. The goal follows from $\mathrm{informationboundaudit}(n, t, w, P)$. □

3 Central collision recurrence

Theorem 3.1. *For every natural number Q and natural number t and natural number r, if* leftadjacentdefect$(Q, t) = \text{true}$*, then if* center$(t + r) = \text{true}$*, then if* center$(t + r + Q) = \text{true}$*, then* nearcollision$(Q, t, r + 1, 2 \cdot (r + 1))$.

Proof. The goal follows from unitdefectforcesnearcollision(Q, t, r). □

Theorem 3.2. *For every natural number T and natural number P and natural number m and natural number t, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq t$, then if* leftadjacentdefect$(m \cdot P, t) = \text{true}$*, then* nearcollision$(m \cdot P, t, P, 2 \cdot P)$.

Proof. The goal follows from periodicunitdefectforcesnearcollision(T, P, m, t). □

Theorem 3.3. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + P \wedge$* centralcollision$(m \cdot P, t, 2 \cdot P)$.

Proof. The goal follows from periodiccentralcollisionwindow(T, P, m, N). □

Theorem 3.4. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then there exists a natural number t such that $N \leq t$ and* centralcollision$(m \cdot P, t, 2 \cdot P)$.

Proof. The goal follows from periodiccentralcollisions(T, P, m, N). □

Theorem 3.5. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentcentralcollision$(m \cdot P, 2 \cdot P)$.

Proof. The goal follows from periodicrecurrentcentralcolumn(T, P, m). □

Theorem 3.6. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) = \text{true}$*, then if* leftadjacentdefect$(Q, b + r) = \text{true}$*, then there exists a natural number s such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge$*leftcollision$(Q, b + s, r + 1)$.

Proof. The goal follows from blackwindowforcesleftcollision(Q, b, r). □

Theorem 3.7. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisionwindow(T, P, m, N). □

Theorem 3.8. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then there exists a natural number t such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisions(T, P, m, N). □

Theorem 3.9. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleftcolumn(T, P, m). □

Theorem 3.10. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

Proof. The goal follows from collisioncodenonzero(Q, t, k). □

Theorem 3.11. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \leq 6$.

Proof. The goal follows from collisioncodebound(Q, t, k). □

Theorem 3.12. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

Proof. The goal follows from collisioncodecolours(Q, t, k). □

Theorem 3.13. *For every natural number Q and natural number t and natural number k,* advancingatdepth(Q, t, k) *if and only if* collisioncode$(Q, t, k) =$ 3 *or* collisioncode$(Q, t, k) = 5$.

Proof. The goal follows from advancingcollisioncodes(Q, t, k). □

Theorem 3.14. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t + 1, k) =$ leftdefect$(Q, t, k + 1)$.

Proof. The goal follows from collisiontransmitsleft(Q, t, k). □

Theorem 3.15. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

Proof. The goal follows from collisiongatesagree(Q, t, k). □

Theorem 3.16. *For every natural number Q and natural number t and natural number k, if* $2 \leq k$*, then* collisioncode$(Q, t, k) = 1 \rightarrow$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t, k-2)$ *and* (collisioncode$(Q, t, k) = 2 \rightarrow$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t+Q, k-2)) \wedge$(collisioncode$(Q, t, k) = 3 \rightarrow$ leftdefect$(Q, t+1, k-1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 4 \rightarrow$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t + Q, k - 2))) \wedge$ (collisioncode$(Q, t, k) = 5 \rightarrow$ leftdefect$(Q, t + 1, k - 1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 6 \rightarrow$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t, k - 2))$).

Proof. The goal follows from collisionrighttransporttable(Q, t, k). □

Theorem 3.17. *For every natural number Q and natural number t and natural number k, if* $0 < k$*, then* collisioncode$(Q, t, k) = 3 \rightarrow$ leftcolour$(Q, t + 1, k-1) = 2$ *and if* collisioncode$(Q, t, k) = 5$*, then* leftcolour$(Q, t+1, k-1) =$ 1.

Proof. The goal follows from advancingrightcolourtable(Q, t, k). □

Theorem 3.18. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t + 1, k) =$ not(leftdefect$(Q, t, k + 1)$) *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatedifferenceiffflip(Q, t, k). □

Theorem 3.19. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k + 1) =$ true*, then* leftdefect$(Q, t + 1, k) =$ false *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from incomingdefecterasediff(Q, t, k). □

Theorem 3.20. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{collisioncode}(Q, t, k) = 0$.

Proof. The goal follows from $\mathrm{erasurehasnocollision}(Q, t, k)$. □

Theorem 3.21. *For every natural number Q and natural number k, if* $0 < k$*, then* $\mathrm{leftcolour}(Q, 0, k) = 0$ *or* $\mathrm{leftcolour}(Q, 0, k) = 2$.

Proof. The goal follows from $\mathrm{seedleftcolours}(Q, k)$. □

Theorem 3.22. *For every natural number Q and natural number k, if* $2 \leq k$*, then* $\mathrm{collisioncode}(Q, 0, k) = 0$.

Proof. The goal follows from $\mathrm{seedleftcollisionzero}(Q, k)$. □

Theorem 3.23. $\mathrm{collisioncode}(1, 1, 2) = 5$.

Proof. The goal follows from seedadvancingcollisionbirth. □

Theorem 3.24. $\mathrm{leftdefect}(3, 0, 3) = \mathrm{true}$ *and* $\mathrm{leftdefect}(3, 1, 2) = \mathrm{false} \wedge \mathrm{leftgatedifference}(3, 0, 2) = \mathrm{true}$.

Proof. The goal follows from seedgateerasure. □

Theorem 3.25. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$*, then if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t + Q) = \mathrm{true}$*, then* $\mathrm{advancingleftcone}(Q, t, k)$.

Proof. The goal follows from $\mathrm{leftdefectblackcone}(Q, t, k)$. □

Theorem 3.26. *For every natural number Q and natural number b and natural number r, if* $0 < r$*, then if* $\mathrm{centerwindowagree}(b, b + Q, r + 1)$*, then if* $\mathrm{center}(b) = \mathrm{true}$*, then if* $\mathrm{leftadjacentdefect}(Q, b + r) = \mathrm{true}$*, then there exists a natural number s such that* $2 \cdot s \leq r + 1$ *and* $s \leq r \wedge \mathrm{advancingleftcollision}(Q, b + s, r + 1)$.

Proof. The goal follows from $\mathrm{blackwindowforcesadvancingcollision}(Q, b, r)$. □

Theorem 3.27. *For every natural number T and natural number P and natural number m and natural number N, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq N$*, then there exists a natural number t such that* $N \leq t$ *and* $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge \mathrm{advancingleftcollision}(m \cdot P, t, P)$.

Proof. The goal follows from periodicadvancingleftcollisionwindow(T, P, m, N). □

Theorem 3.28. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P), *then* recurrentadvancingleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentadvancingleftcolumn(T, P, m). □

Theorem 3.29. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ false *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

Proof. The goal follows from leftdefectfalsecolours(Q, t, k). □

Theorem 3.30. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ true *if and only if* leftcolour$(Q, t, k) = 1$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftdefecttruecolours(Q, t, k). □

Theorem 3.31. *For every natural number Q and natural number t and natural number k, if $t < k$, then* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftcoloursoutsideseedcone(Q, t, k). □

Theorem 3.32. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$, *then $k \leq t + 1$.*

Proof. The goal follows from collisioncodesupport(Q, t, k). □

Theorem 3.33. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatecodenonzero(Q, t, k). □

Theorem 3.34. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) \leq 6$.

Proof. The goal follows from gatecodebound(Q, t, k). □

Theorem 3.35. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

Proof. The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

Theorem 3.36. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, k)$.

Proof. The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

Theorem 3.37. *For every natural number Q and natural number t and natural number k, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3$ *if and only if* $\mathrm{gatecode}(Q, t, k) \leq 3$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 2$ *or* $4 \leq \mathrm{gatecode}(Q, t, k)$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 1$.

Proof. The goal follows from $\mathrm{erasureblackphaseiff}(Q, t, k)$. □

Theorem 3.38. *For every natural number Q and natural number t and natural number k, if $t+1 < k$, then if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 0$.

Proof. The goal follows from $\mathrm{erasureoutsideseedconewhite}(Q, t, k)$. □

Theorem 3.39. *For every natural number Q and natural number t and natural number k, if $2 \leq k$, then if* $\mathrm{advancingatdepth}(Q, t, k)$*, then if* $\mathrm{leftdefect}(Q, t+k, 0) = \mathrm{false}$*, then there exists a natural number j such that* $\mathrm{firstpatherasure}(Q, t, k, j)$.

Proof. The goal follows from $\mathrm{advancingsourcefirsterasure}(Q, t, k)$. □

Theorem 3.40. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number u such that there exists a natural number d such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* $\mathrm{erasureevent}(m \cdot P, u, d)$.

Proof. The goal follows from $\mathrm{periodicerasurewindow}(T, P, m, N)$. □

Theorem 3.41. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P), *then* recurrentsourceerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentsourceerasure(T, P, m). □

Theorem 3.42. noperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from noperiodiffnorecurrentcentralcollision. □

Theorem 3.43. norowcenterperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

Theorem 3.44. problemone *if and only if* norecurrentcentralcollision.

Proof. The goal follows from problemoneiffnorecurrentcentralcollision. □

Theorem 3.45. *If* norecurrentcentralcollision, *then* noperiod.

Proof. Assume h. The goal follows from noperiodofnorecurrentcentralcollision(h). □

Theorem 3.46. *For every natural number T and natural number P, if $0 < P$, then if* norecurrentcentralcollision, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofnorecurrentcentralcollision(T, P). □

Theorem 3.47. *If* norecurrentcentralcollision, *then* problemone.

Proof. Assume h. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision(h). □

Theorem 3.48. *For every natural number b and natural number d and natural number i,* pairphase(wittphi(b), wittxor(wittphi(d), wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$).

Proof. The goal follows from pairphasestep(b, d, i). □

Theorem 3.49. *For every natural number Q and natural number t and natural number k,* alignedphase$(Q, t, t + k + Q) =$ leftphase(Q, t, k).

Proof. The goal follows from alignedphaseleft(Q, t, k). □

Theorem 3.50. *For every natural number Q and natural number t and natural number k,* $\mathrm{leftphase}(Q, t, k) = 0$ *if and only if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{phasezeroiff}(Q, t, k)$. □

Theorem 3.51. *For every natural number Q and natural number t and natural number k,* $\mathrm{leftphase}(Q, t, k) = -1$ *or* $\mathrm{leftphase}(Q, t, k) = 1$ *if and only if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{phaseunitiff}(Q, t, k)$. □

Theorem 3.52. *For every natural number Q and natural number t and natural number k,* $\mathrm{leftphase}(Q, t{+}1, k) = \mathrm{leftphase}(Q, t, k{+}1){+}\mathrm{leftphaseflux}(Q, t, k)$.

Proof. The goal follows from $\mathrm{leftphasebalance}(Q, t, k)$. □

Theorem 3.53. *For every natural number Q and natural number t and natural number k, if* $\mathrm{collisioncode}(Q, t, k) \neq 0$, *then* $\mathrm{leftphaseflux}(Q, t, k) = -2 \cdot \mathrm{leftphase}(Q, t, k + 1)$.

Proof. The goal follows from $\mathrm{collisionphaseflip}(Q, t, k)$. □

Theorem 3.54. *For every natural number Q and natural number t and natural number k, if* $0 < k$, *then if* $\mathrm{advancingatdepth}(Q, t, k)$, *then* $\mathrm{leftphase}(Q, t{+}1, k - 1) = -\mathrm{leftphase}(Q, t, k)$.

Proof. The goal follows from $\mathrm{advancingphaseflip}(Q, t, k)$. □

Theorem 3.55. *For every natural number Q and natural number t and natural number k,* $\mathrm{gatecode}(Q, t, k) = 1 \to \mathrm{leftgatephase}(Q, t, k) = -1$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \to \mathrm{leftgatephase}(Q, t, k) = -1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \to \mathrm{leftgatephase}(Q, t, k) = -1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \to \mathrm{leftgatephase}(Q, t, k) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \to \mathrm{leftgatephase}(Q, t, k) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \to \mathrm{leftgatephase}(Q, t, k) = 1)$.

Proof. The goal follows from $\mathrm{gatephasecases}(Q, t, k)$. □

Theorem 3.56. *For every natural number Q and natural number t and natural number k, if* $\mathrm{erasureevent}(Q, t, k)$, *then* $\mathrm{leftphase}(Q, t + 1, k) = 0$ *and* $\mathrm{leftphaseflux}(Q, t, k) = -\mathrm{leftphase}(Q, t, k{+}1) \wedge (\mathrm{leftphase}(Q, t, k{+}1) = -1 \vee \mathrm{leftphase}(Q, t, k + 1) = 1)$.

Proof. The goal follows from $\mathrm{erasurephasebalance}(Q, t, k)$. □

Theorem 3.57. *For every natural number Q and natural number t and natural number k, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3 \leftrightarrow \mathrm{leftgatephase}(Q, t, k) \cdot \mathrm{leftphase}(Q, t, k+1) = -1$ *and* $\mathrm{leftcolour}(Q, t+1, k) = 0 \leftrightarrow \mathrm{leftgatephase}(Q, t, k) \cdot \mathrm{leftphase}(Q, t, k+1) = 1$.

Proof. The goal follows from $\mathrm{erasureresetproduct}(Q, t, k)$. □

Theorem 3.58. *For every natural number Q and natural number t and natural number k and natural number n,* $\mathrm{pathphase}(Q, t, k, n) = \mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, n)$.

Proof. The goal follows from $\mathrm{pathphasebalance}(Q, t, k, n)$. □

Theorem 3.59. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{leftphase}(Q, t, t + Q) = 1$.

Proof. The goal follows from $\mathrm{seedleftphase}(Q, t)$. □

Theorem 3.60. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{leftphase}(Q, t + (t + Q), 0) = 1 + \mathrm{pathflux}(Q, t, t + Q, t + Q)$.

Proof. The goal follows from $\mathrm{seedcenterphasebalance}(Q, t)$. □

Theorem 3.61. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then* $\mathrm{pathflux}(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

Proof. The goal follows from $\mathrm{periodicseedflux}(T, P, m, N)$. □

Theorem 3.62. $\mathrm{leftphase}(3, 0, 3) = 1$ *and* $\mathrm{leftphase}(3, 1, 2) = 0 \wedge \mathrm{leftphaseflux}(3, 0, 2) = -1 \wedge \mathrm{leftgatephase}(3, 0, 2) = 1$.

Proof. The goal follows from seedphasecancellation. □

Theorem 3.63. *For every natural number Q and natural number t and natural number k and natural number s, if* $\mathrm{rightmostleft}(Q, t, k)$*, then if $s < k$, then if* $\mathrm{centercleanthrough}(Q, t, s)$*, then* $\mathrm{rightmostleft}(Q, t + s, k - s)$.

Proof. The goal follows from $\mathrm{rightmostleftprofile}(Q, t, k, s)$. □

Theorem 3.64. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then* $\mathrm{firstpatherasure}(Q, t, k, k - 2)$ *and* $\mathrm{erasureevent}(Q, t + (k - 1), 0) \wedge (\mathrm{gatecode}(Q, t + (k-1), 0) = 2 \vee \mathrm{gatecode}(Q, t + (k-1), 0) = 5) \wedge \mathrm{center}(t + (k - 1)) = \mathrm{false}$.

Proof. The goal follows from leadingfirsterasure(Q, t, k). □

Theorem 3.65. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number s such that there exists a natural number k such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

Proof. The goal follows from blackwindowforcesleadingsource(Q, b, r). □

Theorem 3.66. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

Proof. The goal follows from periodicleadingsourcewindow(T, P, m, N). □

Theorem 3.67. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P+2 \cdot P \wedge k \leq P \wedge$*leadingadvance$(m \cdot P, t, k) \wedge$firstpatherasure$(m \cdot P, t, k, k-2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

Proof. The goal follows from periodicleadingerasurewindow(T, P, m, N). □

Theorem 3.68. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleadingerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleadingerasure(T, P, m). □

Theorem 3.69. *For every natural number Q and natural number R and natural number t and natural number k,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

Proof. The goal follows from leftphaselagadd(Q, R, t, k). □

Theorem 3.70. *For every natural number Q and natural number R and natural number t and natural number k and natural number n,* pathflux$(Q + R, t, k, n) =$ pathflux$(Q, t, k, n) +$ pathflux$(R, t + Q, k, n)$.

Proof. The goal follows from pathfluxlagadd(Q, R, t, k, n). □

Theorem 3.71. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then* leftcolour$(Q, t+k, 0) = 3 \leftrightarrow$ leftgatephase$(Q, t+(k-1), 0) \cdot$(leftphase$(Q, t, k)+$ pathflux$(Q, t, k, k-1)) = -1$ *and* leftcolour$(Q, t+k, 0) = 0 \leftrightarrow$ leftgatephase$(Q, t+(k-1), 0) \cdot$ (leftphase(Q, t, k) + pathflux$(Q, t, k, k-1)) = 1$.

Proof. The goal follows from leadingphasereset(Q, t, k). □

Theorem 3.72. *For every natural number Q and natural number t, if* leftdefect$(Q, t, 0) =$ false*, then if* leftdefect$(Q, t+1, 0) =$ false*, then* erasureevent$(Q, t, 0)$ *if and only if* gatecode$(Q, t, 0) = 2$ *or* gatecode$(Q, t, 0) = 5$.

Proof. The goal follows from centererasureiffgate(Q, t). □

Theorem 3.73. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* commonwhiteblock(Q, b, n)*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* gatecode$(Q, b+i, 0) = 2$ *or* gatecode$(Q, b+i, 0) = 5$*, then if* gatecode$(Q, b+k, 0) = 2$ *or* gatecode$(Q, b+k, 0) = 5$*, then* gatecode$(Q, b+j, 0) =$ gatecode$(Q, b+i, 0)$.

Proof. The goal follows from whiteblockgateconvex(Q, b, n, i, j, k). □

Theorem 3.74. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* commonwhiteblock(Q, b, n)*, then if* centercleanthrough$(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* erasureevent$(Q, b+i, 0)$*, then if* erasureevent$(Q, b+k, 0)$*, then* erasureevent$(Q, b+j, 0)$ *and* gatecode$(Q, b+j, 0) =$ gatecode$(Q, b+i, 0)$.

Proof. The goal follows from whiteblockerasureconvex(Q, b, n, i, j, k). □

Theorem 3.75. leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ gatecode$(4, 6, 0) = 5$.

Proof. The goal follows from seedleadinggatefive. □

Theorem 3.76. leadingadvance$(7, 7, 4)$ *and* centercleanthrough$(7, 7, 4) \wedge$ gatecode$(7, 10, 0) = 2$.

Proof. The goal follows from seedleadinggatetwo. □

Theorem 3.77. leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

Proof. The goal follows from seedleadingblackreset. □

Theorem 3.78. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase($incomingBit, laterBit,$ transferphase($incomingBi$ *incoming.*

Proof. The goal follows from transferrecover($incomingBit, laterBit, incoming, gate$). □

Theorem 3.79. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase($incomingBit, laterBit, incoming, gate$) = 0 *if and only if incoming =* recoverphase($incomingBit, laterBit, 0, gate$).

Proof. The goal follows from transferkernel($incomingBit, laterBit, incoming, gate$). □

Theorem 3.80. *For every natural number Q and natural number t and natural number k,* leftphase($Q, t{+}1, k$) = transferphase(leftcell($t, k{+}1$), latergate(Q, t, k), leftphase($Q, t, k{+}1$), leftgatephase(Q, t, k)).

Proof. The goal follows from phasetransfer(Q, t, k). □

Theorem 3.81. *For every natural number Q and natural number t and natural number k,* leftphase($Q, t, k{+}1$) = recoverphase(leftcell($t, k{+}1$), latergate(Q, t, k), leftphase($Q, t{+}1, k$), leftgatephase(Q, t, k)).

Proof. The goal follows from phaserecover(Q, t, k). □

Theorem 3.82. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then not* leftgatephase(Q, t, k) = 0 *and* leftphase($Q, t, k{+}1$) = recoverphase(leftcell($t, k{+}1$), latergate(Q, t, k), 0, leftgatephase(Q, t, k)).

Proof. The goal follows from erasurerecovery(Q, t, k). □

Theorem 3.83. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then not not there exists a natural number t such that $N \leq t$ and* erasureevent($m \cdot P, t, 0$) $\wedge$ gatecode($m \cdot P, t, 0$) = g.

Proof. The goal follows from periodicerasureorientation(T, P, m, N, g). □

Theorem 3.84. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then there exists a natural number s such that $s + 1 < k$ and* center$(t + s) =$ true.

Proof. The goal follows from leadingblackbeforearrival(Q, t, k). □

Theorem 3.85. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* leftadjacentdefect$(Q, b+ r) =$ true*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance(Q, b, k).

Proof. The goal follows from blackgapforcesinitialleading(Q, b, r). □

Theorem 3.86. *For every natural number Q and natural number t, if* leadingadvance$(Q, t, 2)$*, then if* centercleanthrough$(Q, t, 2)$*, then* center$(t) =$ true.

Proof. The goal follows from leadingdepthtwoblack(Q, t). □

Theorem 3.87. *For every natural number Q and natural number b and natural number r and natural number g, if $0 < r$, then if* centerwindowagree$(b, b+ Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* erasureevent$(Q, b + r, 0)$*, then if* gatecode$(Q, b + r, 0) = g$*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance$(Q, b, k) \wedge$ firstpatherasure$(Q, b, k, k - 2) \wedge$ gatecode$(Q, b + (k - 1), 0) = g$.

Proof. The goal follows from blackgaperasureorigin(Q, b, r, g). □

Theorem 3.88. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then not not there exists a natural number b such that there exists a natural number k such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k - 1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k - 2) \wedge$ gatecode$(m \cdot P, b + (k - 1), 0) = g$.

Proof. The goal follows from periodicblackerasureorigin(T, P, m, N, g). □

Theorem 3.89. *For every natural number t,* rightcell$(t+1, 1) =$ bxor(center(t), bxor(rightcell$(t, 1)$, right

Proof. The goal follows from rightimportstep(t). □

Theorem 3.90. *For every natural number Q and natural number t, if* center$(t+ Q) =$ center(t)*, then* rightdefect$(Q, t+1, 1) =$ bxor(rightdefect$(Q, t, 1)$, rightimportdifference(Q, t)).

Proof. The goal follows from rightdefectimportbalance(Q, t). □

Theorem 3.91. *For every natural number Q and natural number t, if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then if* rightdefect$(Q, t, 1) =$ true*, then if* rightdefect$(Q, t{+}1, 1) =$ true*, then* rightphase$(Q, t{+}1, 1) = -$ rightphase$(Q, t, 1)$.

Proof. The goal follows from blackresetsurvival(Q, t). □

Theorem 3.92. *For every natural number Q and natural number t, if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then if* rightdefect$(Q, t, 1) =$ false*, then if* rightdefect$(Q, t + 1, 1) =$ true*, then* rightcell$(t, 1) =$ false *and* rightcell$(t + Q, 1) =$ false $\wedge$ rightdefect$(Q, t, 2) =$ true $\wedge$ rightphase$(Q, t{+}1, 1) = -$ rightphase$(Q, t, 2)$.

Proof. The goal follows from blackresetimport(Q, t). □

Theorem 3.93. *For every natural number T and natural number P and natural number m and natural number N and truth value $before$, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then not not there exists a natural number t such that $N \leq t$ and* rightimport$(t) = before \wedge$ rightimport$(t + m \cdot P) =$ not$(before) \wedge$ rightdefect$(m \cdot P, t + 1, 1) =$ not(rightdefect$(m \cdot P, t, 1)$).

Proof. The goal follows from periodicrightimportorientation$(T, P, m, N, before)$. □

Theorem 3.94. *For every natural number Q and natural number t, if* center$(t+ Q) =$ center(t)*, then if* rightdefect$(Q, t, 1) =$ false*, then if* rightdefect$(Q, t + 1, 1) =$ true*, then* rightcell$(t, 1) =$ false *and* rightcell$(t{+}Q, 1) =$ false $\wedge$ rightdefect$(Q, t, 2) =$ true $\wedge$ rightimportdifference$(Q, t) =$ true $\wedge$ rightphase$(Q, t + 1, 1) = (1 - 2 \cdot$ bitvalue(center$(t))) \cdot$ rightphase$(Q, t, 2)$.

Proof. The goal follows from rightheadbirth(Q, t). □

Theorem 3.95. *For every natural number Q and natural number $start$ and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish{-}start)$*, then* rightphase$(Q, finish, 1) =$ centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2)$.

Proof. The goal follows from headimportpathphase$(Q, start, finish)$. □

Theorem 3.96. *For every natural number Q and natural number t, if* leftdefect$(Q, t, 0) =$ false*, then if* leftdefect$(Q, t + 1, 0) =$ false*, then if* center$(t) =$ false*, then* leftphase$(Q, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + 1))) \cdot$ rightphase$(Q, t, 1)$.

Proof. The goal follows from whitecenterphasematching(Q, t). □

Theorem 3.97. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g \wedge$ headimportpath$(m \cdot P, start, b + (k-1)) \wedge$ leftphase$(m \cdot P, b, k) +$ pathflux$(m \cdot P, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ centersign$(start, b + (k - 1) - start) \cdot$ rightphase$(m \cdot P, start, 2)$.

Proof. The goal follows from periodiccoupledcancellation(T, P, m, N, g). □

Theorem 3.98. *For every natural number Q and natural number b and natural number k, if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then* leftphase$(Q, b, k) +$ pathflux$(Q, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ leftgatephase$(Q, b + (k-1), 0)$.

Proof. The goal follows from leadingarrivalphaseeqgatephase(Q, b, k). □

Theorem 3.99. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then* leftgatephase$(Q, finish, 0) =$ centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2)$.

Proof. The goal follows from headimportpathgatephase$(Q, start, finish)$. □

Theorem 3.100. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then not* leftgatephase$(Q, finish, 0) = -($centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2))$.

Proof. The goal follows from headimportpathrejectsnegativegate$(Q, start, finish)$. □

Theorem 3.101. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot$

$P, b, k) \wedge \text{firstpatherasure}(m \cdot P, b, k, k-2) \wedge \text{gatecode}(m \cdot P, b+(k-1), 0) = g \wedge \text{headimportpath}(m \cdot P, \mathit{start}, b+(k-1)) \wedge \text{leftgatephase}(m \cdot P, b+(k-1), 0) = \text{centersign}(\mathit{start}, b+(k-1)-\mathit{start}) \cdot \text{rightphase}(m \cdot P, \mathit{start}, 2)$.

Proof. The goal follows from periodiccoupledgatephasesign(T, P, m, N, g). □

Theorem 3.102. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then* pathflux$(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* pathflux$(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge \text{pathflux}(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

Proof. The goal follows from periodicdoubledseedfluxsplit(T, P, m, N). □

Theorem 3.103. leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ headimportpath$(11, 6, 7) \wedge$centercleanthrough$(11, 6, 1) \wedge$leftgatephase$(11, 7, 0) = 1 \wedge \text{centersign}(6, 1) = 1 \wedge \text{rightphase}(11, 6, 2) = 1$.

Proof. The goal follows from seedcoupledgatesign. □

Theorem 3.104. *For every natural number Q and natural number t and natural number j, if* rightdefect$(Q, t+1, j+1) = \text{true}$*, then* rightdefect$(Q, t, j) =$ true *or* rightdefect$(Q, t, j+1) = \text{true} \vee \text{rightdefect}(Q, t, j+2) = \text{true}$.

Proof. The goal follows from rightdefecthasancestor(Q, t, j). □

Theorem 3.105. *For every natural number Q and natural number t and natural number j and natural number T, if $1 \leq j$, then if $T \leq t$, then if* rightdefect$(Q, t, j) =$ true*, then if* centercleanthrough$(Q, T, t-T)$*, then* rightlineage(Q, t, j, T).

Proof. The goal follows from rightlineageofcleancenter(Q, t, j, T). □

Theorem 3.106. *For every natural number Q and natural number t and natural number k and natural number T, if $1 \leq k$, then if $T \leq t$, then if* leftdefect$(Q, t, k) =$ true*, then if* centercleanthrough$(Q, T, t-T)$*, then* leftlineage(Q, t, k, T).

Proof. The goal follows from leftlineageofcleancenter(Q, t, k, T). □

Theorem 3.107. *For every natural number Q and natural number t and natural number j and natural number T, if $T \leq t$, then if* $\mathrm{rightlineage}(Q, t, j, T)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq j + (t - T) \wedge j \leq i + (t - T) \wedge \mathrm{rightdefect}(Q, T, i) = \mathrm{true}$.*

Proof. The goal follows from $\mathrm{rightlineageatstart}(Q, t, j, T)$. □

Theorem 3.108. *For every natural number Q and natural number t and natural number k and natural number T, if $T \leq t$, then if* $\mathrm{leftlineage}(Q, t, k, T)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq k + (t - T) \wedge k \leq i + (t - T) \wedge \mathrm{leftdefect}(Q, T, i) = \mathrm{true}$.*

Proof. The goal follows from $\mathrm{leftlineageatstart}(Q, t, k, T)$. □

Theorem 3.109. *For every natural number Q and natural number t and natural number j, if* $\mathrm{rightlineage}(Q, t, j, 0)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq Q \wedge i \leq j + t \wedge j \leq i + t \wedge \mathrm{rightdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{rightcell}(Q, i) = \mathrm{true} \wedge \mathrm{rightphase}(Q, 0, i) = 1$.*

Proof. The goal follows from $\mathrm{rightlineageseed}(Q, t, j)$. □

Theorem 3.110. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftlineage}(Q, t, k, 0)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq Q \wedge i \leq k + t \wedge k \leq i + t \wedge \mathrm{leftdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{leftcell}(Q, i) = \mathrm{true} \wedge \mathrm{leftphase}(Q, 0, i) = 1$.*

Proof. The goal follows from $\mathrm{leftlineageseed}(Q, t, k)$. □

Theorem 3.111. *For every natural number Q and natural number t, if $0 < Q$, then there exists a natural number y such that $y \leq t + Q$ and* $\mathrm{rightfront}(Q, t, y)$*.*

Proof. The goal follows from $\mathrm{existsrightfront}(Q, t)$. □

Theorem 3.112. *For every natural number Q and natural number t and natural number y and natural number z, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then $y \leq z + 1$.*

Proof. The goal follows from $\mathrm{rightfrontlower}(Q, t, y, z)$. □

Theorem 3.113. *For every natural number Q and natural number t and natural number y, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if $2 \leq y$, then* $\mathrm{rightdefect}(Q, t + 1, y - 1) = \mathrm{not}(\mathrm{rightcell}(t, y - 1))$*.*

Proof. The goal follows from $\mathrm{rightfrontdescent}(Q, t, y)$. □

Theorem 3.114. *For every natural number Q and natural number t and natural number y and natural number z, if* leftdefect$(Q, t, 0) =$ false*, then if* rightfront(Q, t, y)*, then if* $2 \leq y$*, then if* rightfront$(Q, t + 1, z)$*, then if* $z + 1 \leq y$*, then* $z = y - 1$ *and* rightcell$(t, y - 1) =$ false*.*

Proof. The goal follows from rightfrontlockstep(Q, t, y, z). □

Theorem 3.115. *For every natural number Q and natural number b and natural number k and natural number s, if* rightmostleft(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* $s+1 < k$*, then if* rightdefect$(Q, b+s, 1) =$ true*, then* center$(b + s) =$ true*.*

Proof. The goal follows from leadingheadblackbeforearrival(Q, b, k, s). □

Theorem 3.116. *For every natural number Q and natural number b and natural number k and natural number start, if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k - 1)$*, then if* headimportpath$(Q, start, b + (k - 1))$*, then* $start = b + (k - 2)$*.*

Proof. The goal follows from coupledimportbornatarrival$(Q, b, k, start)$. □

Theorem 3.117. *For every natural number Q and natural number b and natural number k and natural number s, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* $1 \leq s$*, then if* $s+1 < k$*, then there exists a natural number y such that* rightdefect$(Q, b + s, 1) =$ false *and* rightfront$(Q, b + s, y) \wedge 2 \leq y \wedge y \leq k - s$*.*

Proof. The goal follows from coupledrightfrontmirrorat(Q, b, k, s). □

Theorem 3.118. *For every natural number Q and natural number b and natural number k and natural number s, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* $1 \leq s$*, then if* $s + 2 < k$*, then if* rightfront$(Q, b + s, k - s)$*, then* rightcell$(b + s, k - s - 1) =$ false *and* rightfront$(Q, b + (s + 1), k - (s + 1))$*.*

Proof. The goal follows from coupledwhitestaircaseat(Q, b, k, s). □

Theorem 3.119. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if* $0 < P$*, then if* $0 < m$*, then if* centerperiodic(T, P)*, then if* $g = 2$ *or* $g = 5$*, then not not there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that there exists a*

natural number y such that $N \leq start$ and $N \leq b \wedge T \leq b \wedge T \leq start \wedge 2 \leq k \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{gatecode}(m \cdot P, b+(k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b+(k-1)) \wedge (3 \leq k \to start = b+(k-2)) \wedge (3 \leq k \to \mathrm{rightfront}(m \cdot P, b+1, y) \wedge 2 \leq y \wedge y \leq k-1) \wedge \mathrm{rightlineage}(m \cdot P, start, 2, T) \wedge \mathrm{leftlineage}(m \cdot P, b, k, T)$.

Proof. The goal follows from $\mathrm{periodiccoupledimportancestry}(T, P, m, N, g)$. □

Theorem 3.120. $\mathrm{leadingadvance}(14, 5, 3)$ *and* $\mathrm{centercleanthrough}(14, 5, 3) \wedge \mathrm{whitecentergap}(5, 2) \wedge \mathrm{center}(5) = \mathrm{true} \wedge \mathrm{rightfront}(14, 6, 2) \wedge \mathrm{headimportpath}(14, 6, 7)$.

Proof. The goal follows from seedcoupledimportatarrival. □

Theorem 3.121. *For every natural number t and natural number k and natural number n,* $\mathrm{bitvalue}(\mathrm{raybit}(t, k, n)) = \mathrm{bitvalue}(\mathrm{raybit}(t, k, 0)) + \mathrm{rayflux}(t, k, n)$.

Proof. The goal follows from $\mathrm{raybitbalance}(t, k, n)$. □

Theorem 3.122. *For every natural number Q and natural number d and natural number t and natural number p, if $d + p \leq t$, then* $\mathrm{leftphase}(Q + Q, t + (t - d - p), p) = \mathrm{diagonalphase}(Q, d, t) + \mathrm{diagonalflux}(Q, d, t, p)$.

Proof. The goal follows from $\mathrm{diagonalraybalance}(Q, d, t, p)$. □

Theorem 3.123. *For every natural number Q and natural number R and natural number d and natural number t and natural number p,* $\mathrm{diagonalflux}(Q + R, d, t, p) = \mathrm{diagonalflux}(Q, d, t, p) + \mathrm{diagonalflux}(R, d, t + Q, p)$.

Proof. The goal follows from $\mathrm{diagonalfluxlagadd}(Q, R, d, t, p)$. □

Theorem 3.124. *For every natural number Q and natural number z and natural number t, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(Q, z)$*, then if* $\mathrm{settledby}(z+1, t)$*, then* $\mathrm{diagonalphase}(Q, z+1, t) = -1 \vee \mathrm{diagonalphase}(Q, z+1, t) = 1$ *and* $\mathrm{diagonalphase}(Q, z+1, t+Q) = -\mathrm{diagonalphase}(Q, z+1, t) \wedge \mathrm{diagonalphase}(Q + Q, z+1, t) = 0$.

Proof. The goal follows from $\mathrm{diagonalphasedoubles}(Q, z, t)$. □

Theorem 3.125. *For every natural number Q and natural number z and natural number t, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{settledby}(z+2, t)$*, then* $\mathrm{diagonalphase}(Q, z+3, t+1) = -\mathrm{diagonalphase}(Q, z+1, t)$.

Proof. The goal follows from $\mathrm{diagonalresetphase}(Q, z, t)$. □

Theorem 3.126. *For every natural number Q and natural number z and natural number t, if* zerodiagonal(z)*, then if* settledby$(z+2,t)$*, then if* $z+3 \leq t$*, then* diagonalflux$(Q, z+3, t+1, 1) =$ diagonalflux$(Q, z+1, t, 1) + 2 \cdot$ diagonalphase$(Q, z+1, t)$.

Proof. The goal follows from diagonalresetfluxjump(Q, z, t). □

Theorem 3.127. *For every natural number T and natural number P and natural number m and natural number d and natural number t, if* centerperiodic(T, P)*, then if* $d+1 \leq t$*, then if* $T \leq t+(t-d-1)$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t+(t-d-1)+1))) \cdot$rightphase$(m \cdot P + m \cdot P, t+(t-d-1), 1)$.

Proof. The goal follows from periodicdiagonalwhiteheadbalance(T, P, m, d, t). □

Theorem 3.128. *For every natural number T and natural number P and natural number m and natural number d and natural number t and natural number start, if* centerperiodic(T, P)*, then if* $d+1 \leq t$*, then if* $T \leq start$*, then if* headimportpath$(m \cdot P + m \cdot P, start, t+(t-d-1))$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + (t-d-1) + 1))) \cdot$ (centersign$(start, t + (t-d-1) - start) \cdot$ rightphase$(m \cdot P + m \cdot P, start, 2))$.

Proof. The goal follows from periodicdiagonalimportbalance$(T, P, m, d, t, start)$. □

Theorem 3.129. *For every natural number Q and natural number b and natural number k and natural number start and natural number q, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq q$*, then if* $q + 1 < k$*, then* rightfront$(Q, b+1, k-1)$ *and* rightcell$(b+1, q) =$ false $\wedge$ rightcell$(b+1+Q, q) =$ false.

Proof. The goal follows from coupledrightfrontexact$(Q, b, k, start, q)$. □

Theorem 3.130. *For every natural number Q and natural number b and natural number k and natural number start and natural number i, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq i$*, then if* $i + 1 < k$*, then* rightfront$(Q, b+i, k-i)$ *and* rightcell$(b+i, k-i-1) =$ false $\wedge$ rightcell$(b+i+Q, k-i-1) =$ false.

Proof. The goal follows from coupledforcedwhitestaircase$(Q, b, k, start, i)$. □

Theorem 3.131. *For every natural number Q and natural number b and natural number start, if* leadingadvance$(Q, b, 2)$*, then if* centercleanthrough$(Q, b, 2)$*, then if* headimportpath$(Q, start, b+1)$*, then* center$(b) =$ true *and* center$(b+1) =$ false $\wedge$($start < b \wedge$rightdefect$(Q, b, 1) =$ true $\wedge$ rightimportdifference$(Q, b) =$ false $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 1) \vee$ $start = b \wedge$ rightdefect$(Q, b, 1) =$ false $\wedge$ rightcell$(b, 1) =$ false $\wedge$ rightcell$(b + Q, 1) =$ false $\wedge$ rightdefect$(Q, b, 2) =$ true $\wedge$ rightimportdifference$(Q, b) =$ true $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 2))$.

Proof. The goal follows from depthtwocancellationcases$(Q, b, start)$. □

Theorem 3.132. *For every natural number Q and natural number b and natural number k and natural number start, if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then* $start = b+(k-2)$ *or* $k = 2$ *and* $start < b \wedge$rightdefect$(Q, b, 1) =$ true.

Proof. The goal follows from coupledimportbirthordepthtwosurvival$(Q, b, k, start)$. □

Theorem 3.133. *For every natural number Q and natural number b and natural number k and natural number start and natural number q, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* center$(b) =$ true*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $q+2 < k$*, then* rightcell$(b, q) =$ true *and* rightcell$(b+Q, q) =$ true.

Proof. The goal follows from coupledblackrootprefix$(Q, b, k, start, q)$. □

Theorem 3.134. leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ headimportpath$(4, 2, 6) \wedge$rightdefect$(4, 5, 1) =$ true $\wedge$ rightimportdifference$(4, 5) =$ false.

Proof. The goal follows from seeddepthtwosurvival. □

Theorem 3.135. *For every natural number Q and natural number t,* alignedbackground$(Q, t+1) =$ wittphi(alignedbackground(Q, t)).

Proof. The goal follows from alignedbackgroundstep(Q, t). □

Theorem 3.136. *For every natural number Q and natural number t,* $\mathrm{aligneddefect}(Q, t+1) = \mathrm{defectstep}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t))$.

Proof. The goal follows from $\mathrm{aligneddefectstep}(Q, t)$. □

Theorem 3.137. *For every natural number Q and natural number t,* $\mathrm{rowbit}(\mathrm{alignedbackground}(Q, t), 2 \cdot t + Q) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{backgroundtopbit}(Q, t)$. □

Theorem 3.138. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), 2 \cdot (t + Q)) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{aligneddefecttopbit}(Q, t)$. □

Theorem 3.139. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{twopow}(2 \cdot (t + Q)) \leq \mathrm{aligneddefect}(Q, t)$ *and* $\mathrm{aligneddefect}(Q, t) < \mathrm{twopow}(2 \cdot (t + Q) + 1)$.

Proof. The goal follows from $\mathrm{aligneddefectmagnitude}(Q, t)$. □

Theorem 3.140. *For every natural number Q and natural number t and natural number j, if $2 \cdot (t+Q) < j$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{aligneddefectabovetop}(Q, t, j)$. □

Theorem 3.141. *For every natural number Q and natural number t and natural number j, if $j < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{wittrow}(t+Q), j)$.

Proof. The goal follows from $\mathrm{aligneddefectlowprefix}(Q, t, j)$. □

Theorem 3.142. *For every natural number b and natural number d,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittlinear}(d), \mathrm{wittshl}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)), 1))$.

Proof. The goal follows from $\mathrm{defectsteplinear}(b, d)$. □

Theorem 3.143. *For every natural number b and natural number d,* $\mathrm{rowbit}(\mathrm{defectstep}(b, d), 0) = \mathrm{rowbit}(d, 0)$.

Proof. The goal follows from $\mathrm{defectsteplowbit}(b, d)$. □

Theorem 3.144. *For every natural number b and natural number d,* $\mathrm{vala}(\mathrm{defectstep}(b, d)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)))$ *and* $\mathrm{valb}(\mathrm{defectstep}(b, d)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(d$

Proof. The goal follows from $\mathrm{defectstepvalcoordinates}(b, d)$. □

Theorem 3.145. *For every natural number j,* $\mathrm{vala}(\mathrm{wittshl}(1, j+3)) = \mathrm{vala}(\mathrm{wittshl}(1, j))$ *and* $\mathrm{valb}(\mathrm{wittshl}(1, j+3)) = \mathrm{valb}(\mathrm{wittshl}(1, j)) \wedge (\mathrm{vala}(\mathrm{wittshl}(1, j)) = \mathrm{true} \vee \mathrm{valb}(\mathrm{wittshl}(1, j)) = \mathrm{true})$.

Proof. The goal follows from $\mathrm{edgevalcoordinates}(j)$. □

Theorem 3.146. *For every natural number t and natural number k,* $\mathrm{aligneddefect}(\mathrm{twopow}(k), t) = \mathrm{wittxor}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), \mathrm{wittxor}(\mathrm{wittrow}(t), \mathrm{wittshl}(\mathrm{wittrow}(t), \mathrm{twopow}(k) + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{aligneddefectwindow}(t, k)$. □

Theorem 3.147. *For every natural number t and natural number k, if* $t < \mathrm{twopow}(k)$, *then* $\mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 0) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 0) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{false} \wedge \mathrm{vala}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{vala}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{valb}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{valb}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), t + \mathrm{twopow}(k)) = \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{dyadicwindowreadings}(t, k)$. □

Theorem 3.148. *For every natural number T and natural number P and natural number m and natural number t, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then* $\mathrm{rowbit}(\mathrm{wittphi}(\mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P + 1) = \mathrm{rowbit}(\mathrm{crosscarry}(\mathrm{alignedbackground}(m \cdot P, t), \mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P)$.

Proof. The goal follows from $\mathrm{periodiccentercrosscarry}(T, P, m, t)$. □

Theorem 3.149. $\mathrm{aligneddefect}(1, 0) = 5$ *and* $\mathrm{vala}(\mathrm{aligneddefect}(1, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 0)) = \mathrm{true} \wedge \mathrm{aligneddefect}(1, 2) = 93 \wedge \mathrm{vala}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(1, 2), 3) = \mathrm{true} \wedge \mathrm{aligneddefect}(4, 0) = 385 \wedge \mathrm{vala}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(4, 0), 4) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{seedvalcancellation}$. □

Theorem 3.150. *For every natural number d and natural number n and natural number j, if* $j < n$ *or* $n + 2 < j$, *then* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), n) = \mathrm{not}(\mathrm{rowbit}(d, n))$ *and* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), j) = \mathrm{rowbit}(d, j) \wedge \mathrm{vala}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{vala}(d) \wedge \mathrm{valb}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{valb}(d)$.

Proof. The goal follows from $\mathrm{centerflipmaskinvisible}(d, n, j)$. □

Theorem 3.151. *For every natural number t and natural number j,* $\mathrm{rowbit}(\mathrm{observercode}(t), j) = \mathrm{leftcell}(t, j)$.

Proof. The goal follows from $\mathrm{observercodereadout}(t, j)$. □

Theorem 3.152. *For every natural number Q and natural number t,* $\mathrm{observerdefect}(Q, t) = \mathrm{wittxor}(\mathrm{observercode}(t + Q), \mathrm{observercode}(t))$.

Proof. The goal follows from $\mathrm{observerdefectalignment}(Q, t)$. □

Theorem 3.153. *For every natural number Q and natural number t and natural number j,* $\mathrm{rowbit}(\mathrm{observerdefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), t + Q + j)$.

Proof. The goal follows from $\mathrm{observerdefectreadout}(Q, t, j)$. □

Theorem 3.154. *For every natural number t,* $\mathrm{vala}(\mathrm{observercode}(t + 1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{bxor}(\mathrm{vala}(\mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{observerinput}(t)))$ *and* $\mathrm{valb}(\mathrm{observercode}(t+1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{valb}(\mathrm{wittcarry}(\mathrm{observercode}(t))))$.

Proof. The goal follows from $\mathrm{observervalstep}(t)$. □

Theorem 3.155. *For every natural number T and natural number P and natural number m and natural number t, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then* $\mathrm{vala}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t)))), \mathrm{leftdefect}(m \cdot P, t, 1))$ *and* $\mathrm{valb}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))))$.

Proof. The goal follows from $\mathrm{periodicobservervalstep}(T, P, m, t)$. □

Theorem 3.156. *For every natural number T and natural number P and natural number m and natural number t, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$, *then if $T \leq t$, then* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), t + m \cdot P) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), 0) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{periodicobserveredges}(T, P, m, t)$. □

Theorem 3.157. $\mathrm{observerdefect}(2, 0) = 7$ *and* $\mathrm{vala}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(2, 0), 0) = \mathrm{true} \wedge \mathrm{observerdefect}(4, 0) = 24 \wedge \mathrm{vala}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{valb}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(4, 0), 0) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{observerseedzeroval}$. □

4 Cyclotomic observer transport

Theorem 4.1. *For every natural number r and natural number a,* $\mathrm{qdivides}(r, \mathrm{wittlinpow}(r, a))$.

Proof. The goal follows from $\mathrm{qdividesintro}(r, a)$. □

Theorem 4.2. *For every natural number r and natural number d, if* $\mathrm{qdivides}(r, d)$, *then there exists a natural number a such that* $\mathrm{wittlinpow}(r, a) = d$.

Proof. The goal follows from $\mathrm{qdivideselim}(r, d)$. □

Theorem 4.3. *For every natural number r and natural number d and natural number n,* $\mathrm{qdivides}(r, \mathrm{wittshl}(d, n))$ *if and only if* $\mathrm{qdivides}(r, d)$.

Proof. The goal follows from $\mathrm{qdividesshl}(r, d, n)$. □

Theorem 4.4. *For every natural number r and natural number d, if* $0 < r$, *then if* $\mathrm{qdivides}(r, d)$, *then* $\mathrm{vala}(d) = \mathrm{false}$ *and* $\mathrm{valb}(d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{qdividesval}(r, d)$. □

Theorem 4.5. *For every natural number r and natural number d, if* $d \neq 0$, *then if* $\mathrm{qdivides}(r, d)$, *then* $2 \cdot r \leq \mathrm{binarydegree}(d)$.

Proof. The goal follows from $\mathrm{qdividesdegreebound}(r, d)$. □

Theorem 4.6. *For every natural number d, if* $d \neq 0$, *then there exists a natural number r such that* $\mathrm{qdivides}(r, d)$ *and not* $\mathrm{qdivides}(r + 1, d)$.

Proof. The goal follows from $\mathrm{qorderexists}(d)$. □

Theorem 4.7. *For every natural number r and natural number d, if* $d \neq 0$, *then* $\mathrm{qdivides}(r, d)$ *if and only if* $r \leq \mathrm{qorder}(d)$.

Proof. The goal follows from $\mathrm{qdividesifforder}(r, d)$. □

Theorem 4.8. *For every natural number Q and natural number t,* $\mathrm{observersourceflux}(Q, t, 0) = 0$.

Proof. The goal follows from $\mathrm{observersourcefluxzero}(Q, t)$. □

Theorem 4.9. *For every natural number Q and natural number t and natural number n,* $\mathrm{observersourceflux}(Q, t, n{+}1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{ob}$ $n), n + 1))$.

Proof. The goal follows from $\mathrm{observersourcefluxstep}(Q, t, n)$. □

Theorem 4.10. *For every natural number Q and natural number t and natural number n and natural number s,* observersourceflux$(Q, t, n + s) =$ wittxor(wittlinpow$(s,$ observersourceflux$(Q, t, n)),$ wittshl(observersourceflux$(Q, t+ n, s), n)$).

Proof. The goal follows from observersourcefluxappend(Q, t, n, s). □

Theorem 4.11. *For every natural number Q and natural number t,* observerqrank$(Q, t) = t + Q - 2 \cdot$ qorder(observerdefect(Q, t)).

Proof. The goal follows from observerqrankdefinition(Q, t). □

Theorem 4.12. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+1), 1) =$ wittxor(wittlinear(observerdefect$(m \cdot P, t)),$ wittshl(observersource$(m \cdot P, t), 1)$).

Proof. The goal follows from periodicobserverpolynomialstep(T, P, m, t). □

Theorem 4.13. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t+ m \cdot P)),$ wittcarry(observercode$(t))),$ naturalbit(leftdefect$(m \cdot P, t, 1))$).

Proof. The goal follows from periodicobserversource(T, P, m, t). □

Theorem 4.14. *For every natural number T and natural number P and natural number m and natural number t and natural number n, if* centerperiodic(T, P)*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+n), n) =$ wittxor(wittlinpow$(n,$ observerdefect$(m \cdot P, t)),$ observersourceflux$(m \cdot P, t, n)$).

Proof. The goal follows from periodicobserverpolynomialwindow(T, P, m, t, n). □

Theorem 4.15. *For every natural number T and natural number P and natural number m and natural number t and natural number n and natural number r and natural number k, if* centerperiodic(T, P)*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then if $k \leq n+r$, then* qdivides$(k,$ observerdefect$(m \cdot P, t + n))$ *if and only if* qdivides$(k,$ observersourceflux$(m \cdot P, t, n))$.

Proof. The goal follows from periodicobserverwindowdivisibility(T, P, m, t, n, r, k). □

Theorem 4.16. *For every natural number T and natural number P and natural number m and natural number t and natural number r, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $\mathrm{qdivides}(r, \mathrm{observerdefect}(m \cdot P, t))$, *then* $\mathrm{qdivides}(r + 1, \mathrm{observerdefect}(m \cdot P, t+1))$ *if and only if* $\mathrm{qdivides}(r+1, \mathrm{observersource}(m \cdot P, t))$.

Proof. The goal follows from $\mathrm{periodicobserverqgain}(T, P, m, t, r)$. □

Theorem 4.17. *For every natural number Q and natural number t, if* $0 < Q$, *then* $2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t)) \leq t + Q$.

Proof. The goal follows from $\mathrm{observerqorderbound}(Q, t)$. □

Theorem 4.18. *For every natural number T and natural number P and natural number m and natural number t and natural number n, if* $0 < P$, *then if* $0 < m$, *then if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $0 < n$, *then* $\mathrm{observerqrank}(m \cdot P, t + n) < \mathrm{observerqrank}(m \cdot P, t)$ *if and only if* $\mathrm{qdivides}(\mathrm{qorder}(\mathrm{observerdefect}(m \cdot P, t)) + n/2 + 1, \mathrm{observersourceflux}(m \cdot P, t, n))$.

Proof. The goal follows from $\mathrm{periodicobserverqrankdrop}(T, P, m, t, n)$. □

Theorem 4.19. $\mathrm{observerdefect}(5, 2) = 216$ *and* $\mathrm{observerdefect}(5, 3) = 412 \wedge \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 2)) \wedge \neg\, \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 3)) \wedge \mathrm{center}(2) = \mathrm{false} \wedge \mathrm{center}(7) = \mathrm{false} \wedge \mathrm{center}(3) = \mathrm{true} \wedge \mathrm{center}(8) = \mathrm{true}$.

Proof. The goal follows from observerseedqloss. □

Theorem 4.20. *For every natural number r and natural number a and natural number d, if* $\mathrm{wittlinpow}(r, a) = d$, *then* $a \leq d$.

Proof. The goal follows from $\mathrm{qquotientbound}(r, a, d)$. □

Theorem 4.21. *For every natural number d and natural number r, if* $\mathrm{qdivides}(r+1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = r + 1$.

Proof. The goal follows from $\mathrm{qordersearchhit}(d, r)$. □

Theorem 4.22. *For every natural number d and natural number r, if not* $\mathrm{qdivides}(r + 1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = \mathrm{qordersearch}(d, r)$.

Proof. The goal follows from $\mathrm{qordersearchmiss}(d, r)$. □

Theorem 4.23. *For every natural number d,* $\mathrm{qorder}(d) = \mathrm{qordersearch}(d, \mathrm{binarydegree}(d))$.

Proof. The goal follows from qorderdefinition(d). □

Theorem 4.24. *For every natural number r and natural number d, if* qdivides(r, d), *then there exists a natural number a such that $a \leq d$ and* wittlinpow(r, a) = d.

Proof. The goal follows from qdividesboundedwitness(r, d). □

Theorem 4.25. *For every natural number d,* qordersearch($d, 0$) = 0.

Proof. The goal follows from qordersearchzero(d). □

Theorem 4.26. *For every natural number a and natural number y,* wittlinpow(twopow(a), y) = wittxor(wittxor(y, wittshl(y, twopow(a))), wittshl(y, twopow(a)+twopow(a))).

Proof. The goal follows from qfrobeniustwopow(a, y). □

Theorem 4.27. wittlinpow(16, 1) = wittxor(wittxor(1, wittshl(1, 16)), wittshl(1, 32)) *and* wittlinpow(16, wittxor(1, wittshl(1, 16))) = wittxor(1, wittshl(1, 48)).

Proof. The goal follows from qsixteenpage. □

Theorem 4.28. *For every natural number a and natural number y,* qdivides(twopow(a), wittxor(y, witts twopow(a)))).

Proof. The goal follows from qdividespageshifttwopow(a, y). □

Theorem 4.29. *For every natural number a and natural number n and natural number H,* qdivides(twopow(a), wittxor(H, wittxor(pagefold(3·twopow(a), H, n), pagetail(3· twopow(a), H, n)))).

Proof. The goal follows from qdividespagefoldtwopow(a, n, H). □

Theorem 4.30. *For every natural number L and natural number H and natural number p, if $0 < L$, then* page(L, wittcarry(H), p) = wittxor(wittcarry(page(L, H, p)), naturalbit(pa

Proof. The goal follows from pagecarry(L, H, p). □

Theorem 4.31. *For every natural number L and natural number H and natural number n, if $0 < L$, then* pagefold(L, wittcarry(H), n) = carryledger(L, H, n).

Proof. The goal follows from carryfoldeqledger(L, H, n). □

Theorem 4.32. *For every natural number a and natural number H and natural number n,* qdivides(twopow(a), wittxor(wittcarry(H), wittxor(carryledger(3· twopow(a), H, n), pagetail($3 \cdot$ twopow(a), wittcarry(H), n)))).

Proof. The goal follows from carryledgercongruencetwopow(a, H, n). □

Theorem 4.33. *For every natural number T and natural number P and natural number m and natural number t and natural number a, if* centerperiodic(T, P), *then if $T \leq t$, then* qdivides(twopow(a), wittxor(observersource$(m \cdot P, t)$, wittxor(wittxor(carryledger(3·twopow(a), observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger(3·twopow(a), observercode(t), $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))))).

Proof. The goal follows from periodicobserversourceledger(T, P, m, t, a). □

Theorem 4.34. *For every natural number T and natural number P and natural number m and natural number t and natural number r and natural number a, if* centerperiodic(T, P), *then if $T \leq t$, then if $r+1 \leq$* twopow(a), *then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$, *then* qdivides$(r+1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ wittxor(wittxor(carryledger(3·twopow(a), observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger$(3 \cdot$ twopow(a), observercode(t), $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))).

Proof. The goal follows from periodicobserverqgainledger(T, P, m, t, r, a). □

Theorem 4.35. qdivides(16, wittlinpow(16, 1)) *and* wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), witt wittxor(wittlinpow(17, 1), 4)$\wedge\neg$ qdivides(16, wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), wittphi(2)))

Proof. The goal follows from pageresiduenotclosed. □

Theorem 4.36. *For every natural number a and natural number $A0$ and natural number $A1$ and natural number $A2$,* qdivides(twopow(a), wittxor(wittxor(wittxor($A0$, wittshl($A1$ twopow(a))), wittxor(wittxor($A0, A2$), wittshl(wittxor($A1, A2$), twopow(a)))))).

Proof. The goal follows from qblockresidue$(a, A0, A1, A2)$. □

Theorem 4.37. *For every natural number a and natural number $A0$ and natural number $A1$ and natural number $A2$, if $A0 <$* twopow(twopow(a)), *then if $A1 <$* twopow(twopow(a)), *then if $A2 <$* twopow(twopow(a)), *then* qdivides(twopow(a), wittxor(wittxor($A0$, wittshl($A1$, twopow(a)))), wittshl($A2$, twopow$(a)+$ twopow(a)))) *if and only if $A0 = A2$ and $A1 = A2$.*

Proof. The goal follows from qblockchargezeroiff$(a, A0, A1, A2)$. □

Theorem 4.38. *For every natural number a and natural number A, if $A <$* twopow(twopow(a)), *then* wittcarry(wittxor(wittxor(A, wittshl(A, twopow(a)))), wittshl(A, twopow$(a)+$ twopow(a)))) = wittxor(wittxor(wittxor(wittcarry(A), wittshl(wittcarry(A), twopow(a)))), wittshl(witt twopow(a))), wittxor(wittshl(naturalbit(blockendpoint(twopow$(a), A$)), twopow(a))), wittshl(naturalbit twopow(a)))).

Proof. The goal follows from repeatedblockcarry(a, A). □

Theorem 4.39. *For every natural number a and natural number A, if* $A <$ twopow(twopow(a))*, then* qdivides(twopow(a), wittxor(wittcarry(wittxor(wittxor(A, wittshl(A, twopow twopow(a)))), naturalbit(blockendpoint(twopow$(a), A$)))).

Proof. The goal follows from repeatedblockcarrycongruence(a, A). □

Theorem 4.40. *For every natural number a and natural number H and natural number p and natural number A, if* $A <$ twopow(twopow(a))*, then if* page(3·twopow$(a), H, p)$ = wittxor(wittxor(A, wittshl(A, twopow(a))), wittshl(A, twopow(a)+ twopow(a)))*, then* qdivides(twopow(a), wittxor(page(3·twopow(a), wittcarry$(H), p)$, wittxor(naturalbit twopow$(a), H, p)$))))).

Proof. The goal follows from pagecarryrepeated(a, H, p, A). □

Theorem 4.41. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if* $T \leq t$*, then* observersource$(m \cdot P, t)$ = wittxor(wittxor(wittcarry(observerdefect($m \cdot P, t$)), crosscarry(observercode(t), observerdefect($m \cdot P, t$))), naturalbit(leftdefect($m \cdot P, t, 1$))).

Proof. The goal follows from periodicobserversourcecrosscarry(T, P, m, t). □

Theorem 4.42. *There exists a natural number s such that there exists a natural number $s2$ such that* qdivides(16, wittxor$(s, s2)$) *and not* qdivides(16, wittxor(wittphi(s), wittphi$(s2$)

Proof. The goal follows from residuenotcongruence. □

Theorem 4.43. *For every natural number Q and natural number t and natural number j,* rowbit(observerdefect$(Q, t), j)$ = leftdefect(Q, t, j).

Proof. The goal follows from observerdefectleftreadout(Q, t, j). □

Theorem 4.44. *For every natural number r and natural number e and natural number k, if* $e \neq 0$*, then if* qdivides$(r,$ wittshl$(e, k))$*, then* $k + 2 \cdot r \leq$ binarydegree(wittshl(e, k)).

Proof. The goal follows from qfrontdegreebudget(r, e, k). □

Theorem 4.45. *For every natural number n and natural number e and natural number o, if* $e \neq 0$*, then if* twopow$(n) \leq o$*, then if* wittshl$(e, o) <$ twopow$(3 \cdot$ twopow$(n))$*, then not* qdivides(twopow(n), wittshl(e, o)).

Proof. The goal follows from qpageoffsetcharge(n, e, o). □

Theorem 4.46. *For every natural number Q and natural number t and natural number k, if $0 < Q$, then if* rightmostleft(Q, t, k)*, then* $k + 2 \cdot$ qorder(observerdefect$(Q, t)) \leq t + Q$.

Proof. The goal follows from observerfrontqbudget(Q, t, k). □

Theorem 4.47. *For every natural number Q and natural number t and natural number k, if $0 < Q$, then if* rightmostleft(Q, t, k)*, then* $k \leq$ observerqrank(Q, t).

Proof. The goal follows from observerfrontqrank(Q, t, k). □

Theorem 4.48. *For every natural number a and natural number Q and natural number t and natural number p and natural number o, if* twopow$(a) \leq o$*, then if $o < 3 \cdot$* twopow(a)*, then if* rightmostleft$(Q, t, 3 \cdot$ twopow$(a) \cdot p + o)$*, then not* qdivides(twopow(a), page$(3 \cdot$ twopow(a), observerdefect$(Q, t), p))$.

Proof. The goal follows from observerfrontpagecharge(a, Q, t, p, o). □

Theorem 4.49. *For every natural number Q and natural number t, if* leftdefect$(Q, t, 0) =$ false*, then* leftdefect$(Q, t + 1, 0) =$ bxor(leftdefect$(Q, t, 1)$, centergate$(Q, t))$.

Proof. The goal follows from centerdefectcoupling(Q, t). □

Theorem 4.50. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* leftdefect$(m \cdot P, t, 1) =$ centergate$(m \cdot P, t)$.

Proof. The goal follows from periodiccentercoupling(T, P, m, t). □

Theorem 4.51. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then if* center$(t) =$ true*, then* leftdefect$(m \cdot P, t, 1) =$ false.

Proof. The goal follows from periodicblacknoarrival(T, P, m, t). □

Theorem 4.52. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then if* center$(t) =$ false*, then* leftdefect$(m \cdot P, t, 1) =$ rightdefect$(m \cdot P, t, 1)$.

Proof. The goal follows from periodicwhitemirror(T, P, m, t). □

Theorem 4.53. *For every natural number T and natural number P, if* center$(T + P) =$ center(T)*, then if for every natural number t, if $T \leq t$, then* leftdefect$(P, t, 1) =$ centergate(P, t)*, then* centerperiodic(T, P).

Proof. The goal follows from centerperiodicofcoupling(T, P). □

Theorem 4.54. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if* $T \leq t$*, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode($t +$ $m \cdot P$)), wittcarry(observercode(t))), naturalbit(centergate($m \cdot P, t$))).

Proof. The goal follows from periodicobserversourcecentergate(T, P, m, t). □

Theorem 4.55. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if* $T \leq t$*, then if* center$(t) =$ true*, then* observersource$(m \cdot P, t) =$ wittxor(wittcarry(observercode($t +$ $m \cdot P$)), wittcarry(observercode(t))).

Proof. The goal follows from periodicblackobserversource(T, P, m, t). □

Theorem 4.56. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if* $T \leq t$*, then if* center$(t) =$ false*, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode($t +$ $m \cdot P$)), wittcarry(observercode(t))), naturalbit(rightdefect($m \cdot P, t, 1$))).

Proof. The goal follows from periodicwhiteobserversource(T, P, m, t). □

Theorem 4.57. *For every natural number T and natural number P and natural number m and natural number b and natural number k and natural number s, if* centerperiodic(T, P)*, then if* $T \leq b$*, then if* rightmostleft$(m \cdot P, b, k)$*, then if* $s + 1 < k$*, then* leftdefect$(m \cdot P, b + s, 1) =$ false *and* observersource$(m \cdot P, b + s) =$ wittxor(wittcarry(observercode($b + s + m \cdot P$)), wittcarry(observercode($b + s$))).

Proof. The goal follows from periodicfrontgapobserversource(T, P, m, b, k, s). □

Theorem 4.58. *For every natural number T and natural number P and natural number m and natural number b and natural number k, if* centerperiodic(T, P)*, then if* $T \leq b$*, then if* rightmostleft$(m \cdot P, b, k)$*, then* leftdefect$(m \cdot P, b + (k - 1), 1) =$ true *and* center$(b+(k-1)) =$ false $\wedge$ rightdefect$(m \cdot P, b+(k-1), 1) =$ true $\wedge$ observersource$(m \cdot P, b+(k-1)) =$ wittxor(wittxor(wittcarry(observercode($b+$ $(k-1) + m \cdot P$)), wittcarry(observercode($b + (k-1)$))), 1).

Proof. The goal follows from periodicfrontarrivalobserversource(T, P, m, b, k). □

Theorem 4.59. *For every natural number Q and natural number start, if* $\mathrm{center}(start) = \mathrm{false}$*, then if* $\mathrm{center}(start+Q) = \mathrm{false}$*, then if* $\mathrm{center}(start+1) = \mathrm{false}$*, then if* $\mathrm{center}(start+1+Q) = \mathrm{false}$*, then if* $\mathrm{rightcell}(start, 1) = \mathrm{false}$*, then if* $\mathrm{rightcell}(start+Q, 1) = \mathrm{false}$*, then* $\mathrm{leftgatephase}(Q, start+1, 0) = \mathrm{rightphase}(Q, start, 2)$ *and* $\mathrm{leftgatephase}(Q, start+1, 0) = \mathrm{centersign}(start, 1) \cdot \mathrm{rightphase}(Q, start, 2)$.

Proof. The goal follows from $\mathrm{coupledarrivalgatephase}(Q, start)$. □

Theorem 4.60. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), 0) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{aligneddefectbitzero}(Q, t)$. □

Theorem 4.61. *For every natural number T and natural number P and natural number m and natural number b and natural number k, if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq b$, then if* $\mathrm{rightmostleft}(m \cdot P, b, k)$*, then* $\mathrm{leftdefect}(m \cdot P, b + (k-1), 1) = \mathrm{true}$ *and* $\mathrm{centergate}(m \cdot P, b + (k-1)) = \mathrm{true} \wedge \mathrm{leftdefect}(m \cdot P, b + (k-1) + 1, 0) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{periodicarrivalnocrossing}(T, P, m, b, k)$. □